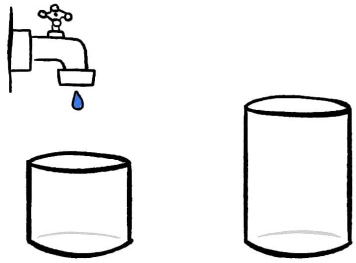
The Jug Problem

How much water would you have
to waste to measure out 2 L?



Allowed Moves

We represent a state of the system by an ordered pair

$$(x, y), \qquad 0 \leq x \leq 4,\ 0 \leq y \leq 7,$$

where x is the amount of water in the 4-litre jug and y is the amount in the 7-litre jug. The following six operations are allowed.

1. **Fill the 4-litre jug from the tap**

$$(x, y) \longrightarrow (4, y) \qquad \text{if } x < 4.$$

2. **Fill the 7-litre jug from the tap**

$$(x, y) \longrightarrow (x, 7) \qquad \text{if } y < 7.$$

3. **Empty the 4-litre jug into the drain**

$$(x, y) \longrightarrow (0, y) \qquad \text{if } x > 0.$$

4. **Empty the 7-litre jug into the drain**

$$(x, y) \longrightarrow (x, 0) \qquad \text{if } y > 0.$$

5. **Pour from the 4-litre jug into the 7-litre jug**

$$(x, y) \longrightarrow (x - t,\ y + t), \qquad t = \min\{x,\ 7 - y\}.$$

6. **Pour from the 7-litre jug into the 4-litre jug**

$$(x, y) \longrightarrow (x + t,\ y - t), \qquad t = \min\{y,\ 4 - x\}.$$

Moves 3 and 4 are **wastes** with costs x and y respectively. The other transitions in the graph (where the states are the nodes) have zero cost.

Invariant condition

The only reachable states are (x is 0 or 4) OR (y is 0 or 7). See the following two figures.

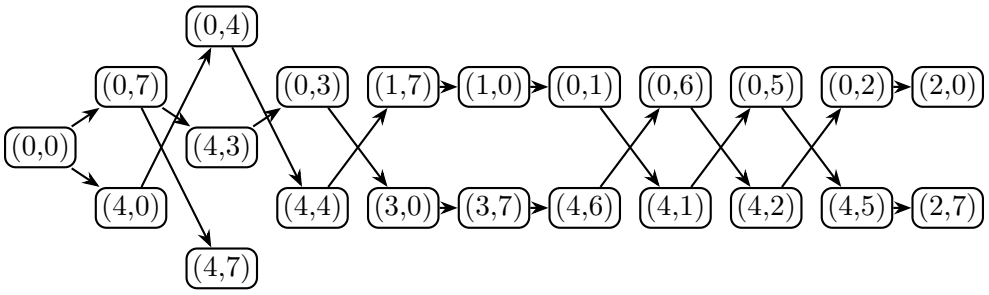


Figure 1: The Breadth First Search (BFS) tree of reachability. Only the $(4,3) \to (0,3)$, $(4,6) \to (0,6)$, and $(1,7) \to (1,0)$ are **wastes** of costs 4, 4, and 7 respectively. One can also do the reverse reachability from $(2,y)$ and $(x,2)$ to argue that there are only two paths.

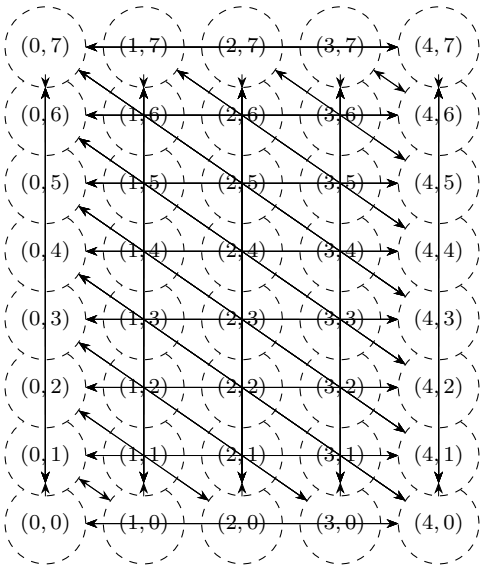


Figure 2: And here is the full reachability graph of 40 states.